THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM THREE ANNUAL EXAMINATION
033 BIOLOGY
MARKING SCHEME NOVEMBER 2025

ASECTION A (16 MARKS)

1. **(10 marks)**

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
A	B	C	D	C	C	B	D	C	B

2. **(05 marks)**

(i)	(ii)	(iii)	(iv)	(v)
B	E	C	D	F

SECTION B (54 MARKS)

3. (a) **Four first aid measures for a person who has suffered from an electric shock**

(i) **Ensure safety:** Before approaching, switch off the power source by unplugging the appliance or turning off the circuit breaker. If you cannot turn off the power, use a dry, non-conducting object (like plastic or wood) to move the source away from the person. **Do not touch the injured person directly** if they are still in contact with the electrical current.

(ii) **Call for emergency medical help immediately:** Dial your local emergency number if the shock is severe, the person is unresponsive, or the burn is from a high-voltage source.

(iii)**Perform CPR if necessary:** If the person is not breathing and has no pulse, begin CPR immediately. Continue chest compressions and rescue breaths until medical help arrives.

(iv)**Cover burns and prevents chilling:** Cover any burned areas with a sterile gauze bandage or a clean cloth, but avoid using a blanket or towel, as fibers can stick to the burns. If the person is conscious, lay them down with their legs elevated, and try to keep them from getting chilled.

(b) **Five reasons why proper waste disposal is important**

(i) **Protects the environment:** Improper waste disposal leads to pollution of air, water, and soil. Harmful chemicals can seep into the ground and contaminate water sources, and waste in landfills releases greenhouse gases like methane.

(ii) **Prevents the spread of diseases:** Waste that is not disposed of correctly can attract pests like rats and insects, which carry diseases. Stagnant waste can become a breeding ground for mosquitoes, which can spread illnesses such as malaria and dengue fever.

(iii) **Conserves natural resources:** Proper waste disposal supports recycling efforts. By recycling materials like paper, metal, and plastic, we reduce the need to extract raw materials, conserving resources and energy.

(iv) **Protects wildlife:** Animals, especially marine life, are harmed by carelessly discarded waste. They can ingest or become entangled in trash, which often leads to injury or death.

(v) **Improves public health:** Ensuring a clean, waste-free environment contributes to overall public well-being. Effective waste management helps maintain hygiene and creates safer living conditions.

4. (a) Malnutrition is a serious condition that occurs when a person's diet does not contain the right amount of nutrients. It can refer to both undernutrition (not getting enough nutrients) and overnutrition (getting more nutrients than the body needs). This can happen when a diet is imbalanced or when there is an inability to properly absorb nutrients.

   (b) Food items and nutrients

   (i) Main nutrient found in each food item:

   - Eggs: Protein

   Irish potatoes: Carbohydrates

   Sunflower seeds: Fats

   (ii) Four functions of the nutrient found in eggs (protein):

   a) Growth and repair: Protein provides the amino acids necessary for building and repairing body tissues, such as muscles, organs, and skin.
   b) Enzyme and hormone production: Many enzymes and hormones are proteins. They regulate body processes, such as metabolism and growth.
   c) Transport of nutrients: Some proteins act as carriers to transport nutrients and other substances throughout the body via the blood.
   d) Immune defense: Proteins are essential for producing antibodies, which are a vital part of the immune system and help fight off infections.

5. (a) Differences between breathing and aerobic respiration

Feature	Breathing	Aerobic Respiration
Process type	A physical/mechanical process involving the movement of air in and out of the lungs.	A biochemical/cellular process involving the breakdown of glucose to produce energy.
Location	Occurs in the lungs and respiratory tract.	Occurs at the cellular level, specifically in the mitochondria.

Energy production	Does not produce energy (ATP).	Produces a large amount of energy (ATP).
Control	A voluntary process, though it can also be involuntary.	An involuntary process regulated by the body's autonomic nervous system.

(b) Factors affecting the rate of gaseous exchange in humans

(i) **Surface area of the alveoli:** A large, moist surface area in the lungs (specifically the alveoli) maximizes the rate of gas diffusion. Diseases like emphysema reduce this surface area, slowing down gaseous exchange.

(ii) **Partial pressure gradient:** The difference in concentration (partial pressure) of a gas between the alveoli and the blood. A steeper gradient, such as that achieved during exercise, increases the rate of exchange.

(iii) **Thickness of the respiratory membrane:** A thinner membrane allows for faster diffusion of gases. Conditions like pulmonary fibrosis increase the thickness, which decreases the rate of exchange.

(iv) **Ventilation rate:** The speed and depth of breathing directly affect the concentration gradients of gases in the lungs. A faster breathing rate, like during exercise, increases gaseous exchange.

(v) **Perfusion:** The blood flow to the lungs (perfusion) affects how quickly gases are carried away from the exchange surfaces. Good ventilation-perfusion matching is crucial for an efficient exchange rate.

6. **(a)Three methods of disease prevention and control**

(i) **Vaccination (Primary Prevention):** Administering vaccines to build immunity against infectious diseases. Vaccines stimulate the body to produce antibodies, offering resistance to specific pathogens.

(ii) **Hygiene and Sanitation (Primordial Prevention):** Practicing good personal hygiene and maintaining clean living conditions. This includes regular handwashing, proper food handling, and sanitizing surfaces to prevent the spread of germs.

(iii) **Health Screening (Secondary Prevention):** Early detection of a disease before symptoms appear. Regular screenings can help in early diagnosis and prompt treatment, preventing the progression of a disease.

(b) **Differentiating between antiseptic and disinfectant**

- **Antiseptic:** A substance used on living tissue, such as skin, to kill or inhibit the growth of microorganisms. It is less concentrated and gentler to avoid harming living cells.
  - **Example:** Hydrogen peroxide.

- **Disinfectant:** A chemical agent used on non-living objects and surfaces to destroy microorganisms. It is typically stronger and potentially harmful to living tissue.
  - **Example:** Bleach.

(c) **Three examples of zoonotic diseases**

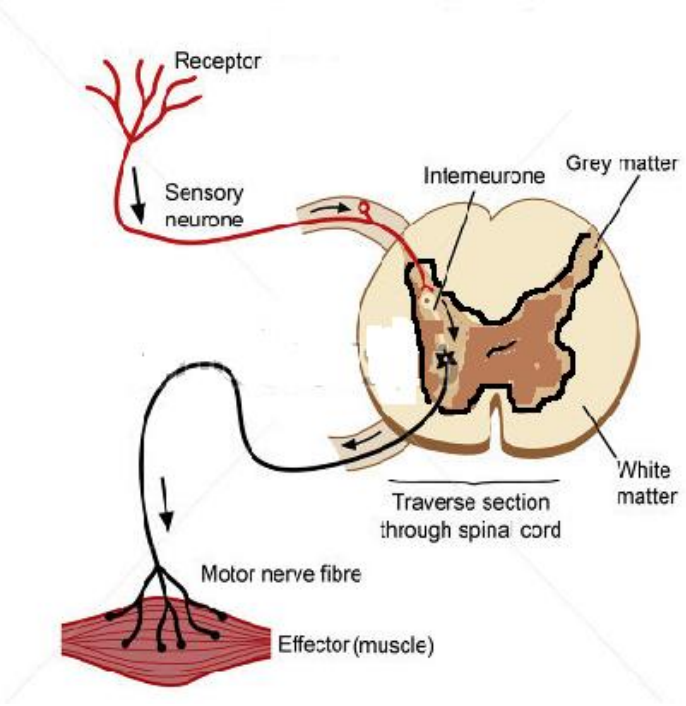
(i) **Rabies:** Transmitted through the bite of an infected animal, like a raccoon or bat.

(ii) **Salmonellosis:** A foodborne illness caused by bacteria often found in contaminated animal products like eggs and poultry.

(iii)**Influenza (e.g., Avian or Swine Flu):** Some strains of the influenza virus originate in animals and can spread to humans.

7. **(a)** A reflex arc is the neural pathway that mediates a reflex action. It involves five main components: a sensory receptor, a sensory neuron, an integration center (usually in the spinal cord), a motor neuron, and an effector. This pathway allows for rapid, involuntary responses to stimuli without involving conscious thought.

Diagram of a Reflex Arc (Pain Withdrawal Reflex):



- **Receptor:** Detects a stimulus (e.g., pain from a sharp object).
- **Sensory Neuron:** Transmits the nerve impulse from the receptor to the spinal cord.
- **Spinal Cord (Integration Center):** The impulse travels here. A relay neuron links the sensory and motor neurons.
- **Motor Neuron:** Transmits the impulse from the spinal cord to the effector.
- **Effector:** A muscle or gland that carries out the response (e.g., muscle contracting to pull the limb away).

(b) Role of the two hormones produced by the pancreas

The two main hormones produced by the pancreas are insulin and glucagon, which work together to regulate blood sugar levels.

- **Insulin:** Produced by the beta cells in the islets of Langerhans, insulin is released when blood glucose levels are high, typically after a meal. It signals body cells (muscle, fat, and liver) to absorb glucose from the blood for energy or storage, thereby lowering blood sugar.

- **Glucagon:** Produced by the alpha cells, glucagon is released when blood glucose levels are low. It signals the liver to convert stored glycogen back into glucose and release it into the bloodstream, raising blood sugar levels.

8. (a) **Over-heating is adjusted through:**

(i) **Sweating:** this is the production of water fluid containing dissolved salt from sweat glands in the skin. Evaporation of sweat brings about a cooling effect. Sweating is associated with loss of body water (dehydration). So, in hot climate a person must drink a lot of water.

(ii) **Panting**: Animals of the dog family have sweat glands only in the pads of their paws. Their main method of losing heat is to pant rapidly with the tongue hanging outside. This causes evaporation from the tongue, mouth and lungs which cool the body.

(iii) **Vasodilatation:** during vasodilatation superficial capillaries below the skin become wide so that more blood flows to the skin surface and heat is lost through the skin surface; that's why the skin feels hot and looks red.

(iv) **Relaxation of hair erector muscles:** hence hair lies more or less flat against the skin, so heat is easily lost by radiation and convection.

(v) **Reduced rate of metabolism:** rate of respiration and other metabolism is reduced and this reduces heat production. *(Any five, 01 @ =05 marks)*

(vi) **Behavioral** pattern: this includes wearing thin and light-colored cloths, taking cold drinks, cold showers, resting in shade and using fan or air-conditioners.

Over-cooling is controlled through:

(i) **Increase in metabolic rate:** increase in the rate of respiration and other metabolic activities lead to heat generation. *(Any four, 01 @ =04 marks)*

(ii) **Vasoconstriction**: this involves the narrowing of capillaries below the skin so that less blood flows through the skin and less heat gets lost through radiation. The skin looks pale during vasoconstriction

(iii) **Shivering**: this is heat-production process in which skeletal muscles are stimulated into uncontrolled activity. These muscles use up the metabolites and produce heat.

(iv) **Contraction of hair erector muscles**: hairs are raised to make a thick coat thus trapping more air as an insulating layer. The contracted hair muscles in naked skin appear as "goose pimples." Also, the upright hairs prevent the cold winds from reaching the skin. Birds achieve a similar effect by means of muscles, which makes their hair fluff out.

(v) **Behavioral pattern:** this includes vigorous exercise, wearing jackets, using heaters, crouching, and warm showers.

$01^{1}/_{2}$ marks @ total = (09 marks)

SECTION C (30 MARKS)

9. Introduction: (**1½ Marks)**

Advantages of Angiosperms

1. **Agriculture.** Agriculture depends on angiosperms. Angiosperms provide human food. Example; plants belonging to grass family (grains), are the most important food stocks.

2. **Livestock.** Plants belonging to grass family are the world number one feeds of livestock. Many types of grassland in Africa are either grazed by cattle or wild animals. Also almost all feeds consumed by human are also used by domestic animals as food.

3. **Wood.** Angiosperms provide different varieties of wood for furniture, paper and building materials. Carpentry and masonry depends largely on angiosperms for both soft and hard wood.

4. **Textile.** Different varieties of fibers including cotton, flax, and hemp are important raw materials for textile industry. Cotton is the word first class material for clothes.

5. **Medicine.** Many organic drugs are coming from angiosperm herbs. They are either industrial made or can be utilized in its raw form. Today it is believed that herbs form the best medicine when used in their fresh form than when industrial made.

6. **Floriculture.** Flowers from angiosperms have great social significance as they are used in different kinds of ceremony. Roses for instance are the most sold flowers and they are highly purchased and used in wedding and burial ceremonies.

7. **Forestry.** Forestry is the field of study whose base depends on both angiosperms and gymnosperms. Many forests form green belts that habit different varieties of plants.

8. **Ecology.** Angiosperms provide a wide range of habitats for different species of organisms. They also serve as primary producers of energy.

9. **Climate.** Angiosperms are used to modify climate. Green belts are established in order to absorb carbon and reduce global warming.

10. **Tourism.** Many sites of tourism are plants oriented. Ecotourism for instance depends largely on different plant species that host different kinds of land animals. **(@2½ Marks)**

Disadvantages of Angiosperms

1. **Toxins.** Some plants are poisonous when eaten by human and other animals. They cause death to human and other living organisms. Some cassava varieties are a good example. They have cyanide acid that inhibits cellular respiration.

2. **Drugs.** Some angiosperm varieties are drugs that can be abused when taken. Examples are marijuana, cocaine, tobacco and some caffeine varieties.

3. **Weeds.** Some angiosperm varieties are bad weeds; they reduce crop yields. Some weeds are parasites. Example, Cuscuta kilimanjari is parasitic to coffee plantations **(@1½ Marks)**

Conclusion: (1½ Marks)

10. (a) Four ways a plant is adapted for transporting water:

1. **Root Hairs for Absorption:** Plants have extensive root systems with numerous tiny root hairs, which greatly increase the surface area for absorbing water and dissolved minerals from the soil through osmosis.

2. **Cohesion and Adhesion:** Water molecules are attracted to each other (cohesion) and to the walls of the xylem vessels (adhesion). This allows for the formation of a continuous, unbroken column of water from the roots to the leaves.

3. **Transpirational Pull:** Water evaporating from the leaves (transpiration) creates a negative pressure (tension) that pulls the entire water column upwards through the xylem. This "pulling" force is the primary driver of water transport in tall plants.

4. **Xylem Structure:** Xylem vessels are hollow, continuous tubes made of dead cells with lignified walls. This structure provides mechanical support and creates a low-resistance pathway for water to move efficiently against gravity. **(@2½ Marks)**

(b) Five features making the human heart an efficient pump:

1. **Four Chambers:** The heart is divided into four chambers: two atria and two ventricles. This separation allows for a double circulatory system, where oxygenated blood is kept separate from deoxygenated blood, ensuring that body tissues receive a constant supply of highly oxygenated blood.

2. **Valves:** The heart has one-way valves (tricuspid, pulmonary, mitral, and aortic) that prevent the backflow of blood. These valves ensure that blood flows in a single, correct direction through the heart and into the circulatory system.

3. **Thick, Muscular Walls:** The heart wall, especially the left ventricle, has a thick layer of muscle (myocardium) that can generate strong contractions. The left ventricle is the thickest because it must pump oxygenated blood to the entire body.

4. **Coordinated Electrical System:** The heart's natural pacemaker (sinus node) sends electrical signals that cause the atria and ventricles to contract in a regular, coordinated rhythm. This ensures efficient pumping of blood with each heartbeat.

5. **Coronary Circulation:** The heart muscle itself has its own blood supply via the coronary arteries, ensuring that its cells are constantly supplied with oxygen and nutrients, which is necessary for it to function continuously. **(@2½ Marks)**

11. **(a) Five events in a typical menstrual cycle:**

1. **Menstruation:** The cycle begins with the shedding of the uterine lining (endometrium) if a fertilized egg has not implanted. This results in menstrual bleeding and typically lasts 3–7 days.

2. **Follicular Phase:** Following menstruation, the pituitary gland releases follicle-stimulating hormone (FSH). This stimulates follicles in the ovaries to develop, with one

maturing into an egg. The follicles produce estrogen, which causes the uterine lining to thicken.

3. **Ovulation:** Around the midpoint of the cycle, a surge in luteinizing hormone (LH) triggers the release of the mature egg from the ovary. This is the most fertile time of the cycle.

4. **Luteal Phase:** After the egg is released, the ruptured follicle becomes the corpus luteum, which produces progesterone. Progesterone further thickens the uterine lining, preparing it for possible pregnancy.

5. **Breakdown of Corpus Luteum:** If pregnancy does not occur, the corpus luteum degenerates, causing progesterone levels to drop. This drop triggers the breakdown of the uterine lining, and menstruation begins again, starting a new cycle. **(@1 Marks)**

(b) Five factors contributing to increased STIs among young people:

1. **Biological Susceptibility:** Young women are more biologically vulnerable to some STIs because their cervixes are not fully developed and are more susceptible to infection.

2. **Risk-Taking Behaviors:** Younger people are more likely to engage in sexual risk-taking behaviors, including having multiple partners and not using protection consistently.

3. **Inconsistent or No Condom Use:** Inconsistent use of condoms, often driven by the use of other forms of birth control that do not protect against STIs, is a major contributing factor.

4. **Substance Abuse:** The use of alcohol or drugs can lower inhibitions and impair judgment, leading to increased rates of unprotected or risky sexual activity.

5. **Lack of Knowledge and Communication:** Many young people lack accurate information about STI prevention, transmission, and testing. They may also be hesitant to discuss sexual health with partners, healthcare providers, or parents. **(@2 Marks)**